Reporting the Orientation of Planes and Lineations

Planes

There are two ways to specify the orientation of a plane
(1) strike and dip: 030/40 SE
(2) dip and dip direction: 40/120

• When specifying the orientation of the plane as strike and dip
(a) the strike should be specified as a **three** figure number, and be stated first; it is also good practice to specify it as a number between 000° and 180°
(b) the amount of dip should be specified as a **one or two** figure number
(c) the general direction of dip should be stated

• When specifying the orientation of the plane as dip and dip direction
(a) the amount of dip should be specified as a **one or two** figure number, and be stated first;
(b) the dip direction should be specified as a **three** figure number, and it must not be converted to a number between 000° and 180°

• Either way of specifying the orientation of a plane are fine, and each will be encountered in the practical classes so that you become accustomed to them both. Preference for one over the other generally depends upon the type of compass clinometer used to make the measurement. The Silva compass clinometers are better suited to making measurements of strike and dip.

Lineations

There are two ways to specify the orientation of a line
(1) plunge and plunge direction: 23/060
(2) pitch on a specified plane: 37 NE on 030/40 SE

• When specifying the orientation of the line as plunge and plunge direction
(a) the amount of plunge should be specified as a **one or two** figure number, and be stated first;
(b) the plunge direction should be specified as a **three** figure number, and it must not be converted to a number between 000° and 180°

• When specifying the orientation of the line as a pitch on a specified plane
(a) the amount of pitch should be specified first and be accompanied by an indication of which end of the strike line it has been measured from
(b) the orientation of the plane on which the pitch was measured must also be stated (as strike and dip or dip and dip direction)

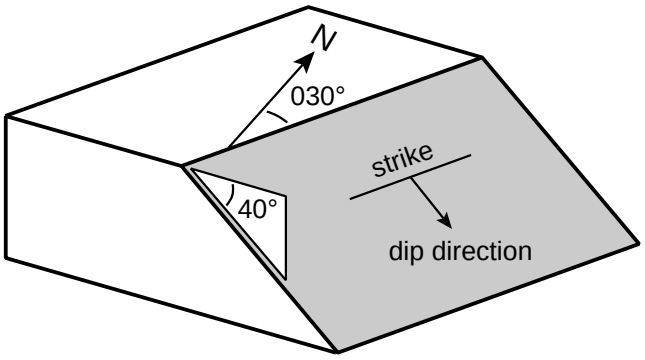
• Linear data should ultimately be expressed as plunge and plunge direction, that is, when specifying the orientation of a line as a pitch on a plane (as one might do in a number of measuring situations), the pitch must subsequently be converted into a plunge and plunge direction using the stereonet.
• Note that ‘pitch’ is formally the same as ‘apparent dip’

When specifying the orientation of a plane or line ALWAYS specify what the plane or line is.

Some conventional shorthand:
(1) for bedding or cleavage surfaces use the letter S, *e.g.*, S_0 bedding, S_1 first cleavage, S_2 second cleavage, *etc.*: S_1 065/42 SE
(2) for lineations the nature of the lineation must be specified, *e.g.*,
• S_1/S_0 intersection lineation of first cleavage on bedding: S_1/S_0 23/246 (Z)
• L_1 mineral elongation or stretching lineation (specified) formed in the first deformation: L_1 23/328
• F_1 the orientation of a fold axis belonging to a fold formed in the first deformation: F_1 23/246 (Z)

SPECIFYING THE ORIENTATION OF PLANES AND LINES

Planes



strike / dip

strike 030° ; dip 40° → 030 / 40 SE

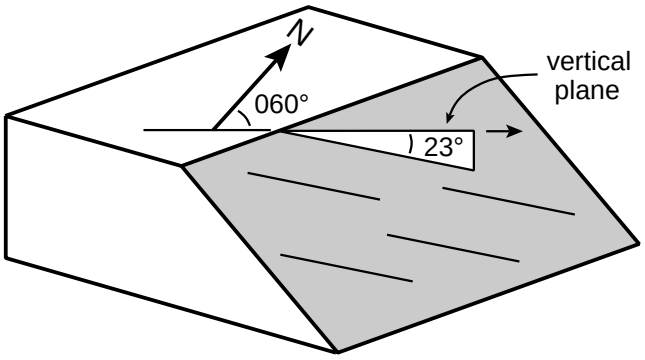
dip / dip direction

dip direction 030 + 90 = 120° → 40 / 120

The horizontal line (contour) which can be drawn on a dipping surface is referred to as a *strike line* (= *stratum contour*); the compass-bearing of this line is referred to as the ***strike*** of the surface.

The ***dip*** of a plane is the angle which that plane makes with the horizontal as measured at 90° to the strike.

Lines

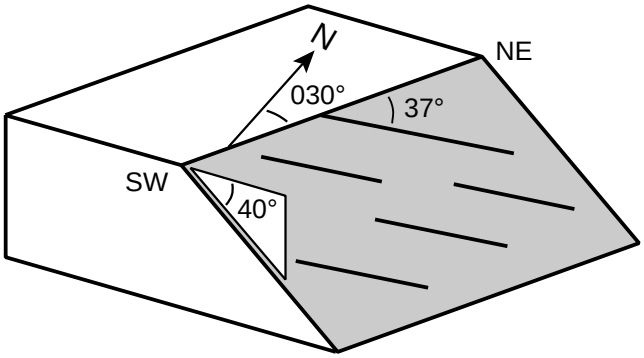


plunge / plunge direction

plunge 23° ; plunge direction 060°
→ 23 / 060

The ***plunge*** of a line is the angle between that line and the horizontal.

The ***plunge direction*** is the compass-bearing towards which the line is tilted.



pitch on a plane

pitch 37°NE ; plane 030 / 40 SE
→ 37 NE on 030 / 40 SE
which is the same as
→ 143 SW on 030 / 40 SE

Pitch is the angle between the strike line and the lineation as measured by lying a protractor on the plane containing the lineation. Note that if the plane in question is a vertical cliff-face and the lineation the intersection of bedding on that cliff-face, then pitch is nothing more than the apparent dip of bedding on that cliff-face